

Boss Challenge — Paper 51 (Class 7)

Heat | Nutrition in Plants | Fractions & Decimals | Data Handling

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A measure of how hot or cold a body is, read on a thermometer, is called its:
(A) temperature
(B) pressure
(C) humidity
(D) heat
2. The process by which green plants make their own food using sunlight, water and carbon dioxide is:
(A) transpiration
(B) photosynthesis
(C) digestion
(D) respiration
3. Among the four fractions $\frac{3}{4}$, $\frac{2}{3}$, $\frac{1}{2}$ and $\frac{5}{12}$, which one has the greatest value?
(A) $\frac{5}{12}$
(B) $\frac{2}{3}$
(C) $\frac{1}{2}$
(D) $\frac{3}{4}$
4. The arithmetic mean (average) of a set of values is found by:
(A) adding all the values and dividing by how many there are
(B) subtracting the smallest from the largest
(C) choosing the most common value
(D) picking the value in the middle
5. A clinical thermometer reads from 35 °C to about 42 °C because that range covers:
(A) the boiling point of water
(B) the human body's temperatures
(C) room temperature only
(D) the temperature of ice
6. Which green pigment, present in a plant's leaves, traps sunlight so that photosynthesis can take place?
(A) starch
(B) haemoglobin
(C) chlorophyll
(D) cellulose

7. What is $\frac{2}{3} + \frac{1}{4}$?

- (A) $\frac{11}{12}$
- (B) $\frac{3}{12}$
- (C) $\frac{11}{7}$
- (D) $\frac{3}{7}$

8. The mean of 4, 8, 6, 10 and 7 is:

- (A) 8
- (B) 7
- (C) 35
- (D) 6

9. Left to itself, heat moves spontaneously out of a hotter object and into one that is at a:

- (A) equal temperature
- (B) a vacuum only
- (C) higher temperature
- (D) lower temperature

10. Tiny pores on the surface of a leaf through which gases enter and leave are called:

- (A) stomata
- (B) chloroplasts
- (C) roots
- (D) veins

11. What is $\frac{3}{5}$ of 200?

- (A) 40
- (B) 80
- (C) 120
- (D) 60

12. In the data 5, 3, 8, 3, 9, 3, 6, the value that occurs most often (the mode) is:

- (A) 3
- (B) 9
- (C) 5
- (D) 6

13. Heat travelling through a solid metal rod, particle to particle without the particles moving along, is:

- (A) radiation
- (B) convection
- (C) evaporation
- (D) conduction

14. Plants that make their own food by photosynthesis are called:
- (A) decomposers
 - (B) parasites
 - (C) autotrophs
 - (D) heterotrophs
15. What is $\frac{1}{2} \times \frac{4}{5}$?
- (A) $\frac{5}{8}$
 - (B) $\frac{4}{7}$
 - (C) $\frac{4}{5}$
 - (D) $\frac{2}{5}$
16. To find the median of a data set, you first:
- (A) take the largest value
 - (B) pick the most frequent value
 - (C) add them all and divide
 - (D) arrange the values in order, then take the middle one
17. Heat reaching us from the Sun across empty space travels by:
- (A) radiation
 - (B) conduction
 - (C) boiling
 - (D) convection
18. Animals and most non-green organisms, which cannot make their own food, are called:
- (A) chlorophyll-bearing
 - (B) autotrophs
 - (C) heterotrophs
 - (D) producers
19. What is $\frac{3}{4} \div \frac{1}{2}$?
- (A) $\frac{3}{2}$ (or $1\frac{1}{2}$)
 - (B) $\frac{2}{3}$
 - (C) $\frac{3}{8}$
 - (D) $\frac{3}{6}$
20. The median of the ordered data 2, 5, 7, 9, 11 is:
- (A) 9
 - (B) 5
 - (C) 6.8
 - (D) 7
21. Materials such as copper and aluminium that let heat pass through them easily are called:
- (A) conductors
 - (B) liquids
 - (C) insulators
 - (D) radiators of light

22. Cuscuta (amarbel), a leafless yellow climber that takes food from the plant it twines around, is a:
- (A) autotroph
 - (B) parasite
 - (C) insectivorous plant
 - (D) saprotroph
23. Expressed in decimal form, the fraction seven-tenths ($\frac{7}{10}$) is equal to:
- (A) 1.7
 - (B) 0.07
 - (C) 0.7
 - (D) 7.0
24. The range of the data 12, 7, 20, 5, 16 is:
- (A) 15
 - (B) 25
 - (C) 5
 - (D) 20
25. A cooking pot is metal but its handle is often plastic or wood because those materials are:
- (A) heavier than metal
 - (B) good conductors of heat
 - (C) poor conductors (insulators) of heat
 - (D) magnetic materials
26. The pitcher plant traps and digests insects mainly to obtain the nutrient:
- (A) carbon
 - (B) water
 - (C) sunlight
 - (D) nitrogen
27. What is $0.25 + 0.4$?
- (A) 0.65
 - (B) 0.65 is wrong, 6.5
 - (C) 0.29
 - (D) 0.45
28. A bar graph is most useful for:
- (A) showing how a whole is divided into parts
 - (B) plotting the exact path of a moving car
 - (C) comparing the sizes of different categories at a glance
 - (D) listing values with no comparison

29. On a hot sunny day light-coloured clothes feel more comfortable than dark ones because light colours:

- (A) are always thinner cloth
- (B) block sunlight completely
- (C) reflect most of the heat and absorb less
- (D) absorb most of the heat

30. Fungi such as bread mould, which feed on dead and decaying matter, are called:

- (A) autotrophs
- (B) insectivores
- (C) parasites
- (D) saprotrophs

31. Two rainfall gauges read 1.6 cm and 0.85 cm. The total rainfall recorded is:

- (A) 0.75 cm
- (B) 2.45 cm
- (C) 9.6 cm
- (D) 2.41 cm

32. In a pictograph, one symbol of a tree stands for 10 saplings planted. A row showing $4\frac{1}{2}$ tree symbols means:

- (A) 45 saplings
- (B) 4 saplings
- (C) 40 saplings
- (D) 450 saplings

33. The warm breeze that blows from the sea toward the land during the daytime is the:

- (A) monsoon wind
- (B) sea breeze
- (C) cyclone
- (D) land breeze

34. In the lichen, a fungus and an alga live together helping each other. This close partnership is called:

- (A) competition
- (B) symbiosis
- (C) parasitism
- (D) predation

35. What is 0.6×0.3 ?

- (A) 0.9
- (B) 0.18
- (C) 1.8
- (D) 0.018

36. A coin is tossed once. The probability of getting heads is:

- (A) 2
- (B) 0
- (C) $\frac{1}{2}$
- (D) 1

37. Water and most liquids are heated by convection, in which the heated liquid:

- (A) turns solid as it warms
- (B) passes heat without moving
- (C) stays at the bottom and gets denser
- (D) rises and cooler liquid sinks to take its place

38. How do Rhizobium bacteria, housed in the root nodules of legume crops, benefit their host plant?

- (A) digesting insects for the plant
- (B) fixing nitrogen from the air into a usable form
- (C) making the plant's chlorophyll for it
- (D) carrying water up from the roots

39. What is $4.5 \div 0.5$?

- (A) 9
- (B) 0.9
- (C) 90
- (D) 2.25

40. Rolling an ordinary die once, the probability of getting a number greater than 4 is:

- (A) $\frac{1}{6}$
- (B) $\frac{2}{6}$ (= $\frac{1}{3}$)
- (C) $\frac{3}{6}$
- (D) $\frac{4}{6}$

41. Two equal cups of tea start at 80 °C. After ten minutes the open cup is 50 °C and the lidded cup is 64 °C. The open cup lost how many more degrees?

- (A) 14 °C more
- (B) 30 °C more
- (C) 16 °C more
- (D) 46 °C more

42. Why do farmers add manure or fertilisers to fields and rotate crops?

- (A) to give the plants their green colour
- (B) to trap insects in the soil
- (C) to block sunlight from the roots
- (D) to replace the soil nutrients, especially nitrogen, that crops use up

43. The mixed number $2\frac{3}{4}$ written as an improper fraction is:

- (A) $\frac{5}{4}$
- (B) $\frac{23}{4}$
- (C) $\frac{11}{4}$
- (D) $\frac{6}{4}$

44. Forty children chose a favourite fruit; the table shows mango 18, apple 12, banana 10. The most popular fruit (the mode) is:

- (A) mango
- (B) apple
- (C) all are equal
- (D) banana

45. Over four days the noon temperatures were 31, 33, 30 and 34 °C. The mean (average) noon temperature was:

- (A) 128 °C
- (B) 31 °C
- (C) 32 °C
- (D) 33 °C

46. During photosynthesis the gas a green leaf takes IN from the air is:

- (A) carbon dioxide
- (B) nitrogen
- (C) oxygen
- (D) hydrogen

47. Which decimal is the smallest: 0.5, 0.45, 0.405, 0.54?

- (A) 0.45
- (B) 0.54
- (C) 0.405
- (D) 0.5

48. Over a week a plant's daily growth (mm) was 3, 5, 4, 6, 2, 4, 4. Its mean daily growth was:

- (A) 3.5 mm
- (B) 28 mm
- (C) 4 mm
- (D) 6 mm

49. A liquid thermometer works because the liquid inside it, on being heated:

- (A) contracts and falls down the tube
- (B) changes into a gas
- (C) expands and rises up the thin tube
- (D) changes colour with heat

50. The food made by green leaves is mainly stored in the plant in the form of:
- (A) starch
 - (B) water
 - (C) oxygen
 - (D) chlorophyll
51. A $\frac{3}{4}$ -litre bottle of juice is shared equally among 3 friends. Each friend gets:
- (A) $\frac{1}{4}$ litre
 - (B) $\frac{3}{12}$ left over
 - (C) $\frac{3}{4}$ litre
 - (D) $\frac{1}{3}$ litre
52. For which purpose is a double bar graph the most suitable kind of graph to draw?
- (A) display data that has no categories
 - (B) show the parts of a single whole
 - (C) plot one quantity changing over time
 - (D) compare two sets of data side by side for each category
53. A maximum-minimum thermometer is useful to a weather station because in one reading it tells the:
- (A) humidity of the air
 - (B) highest and lowest temperature of the day
 - (C) average temperature directly
 - (D) wind speed at noon
54. If iodine solution is dropped on a leaf that has photosynthesised, the leaf turns blue-black, which shows the presence of:
- (A) protein
 - (B) chlorophyll
 - (C) starch
 - (D) oxygen
55. What is $5 - 2.75$?
- (A) 2.75
 - (B) 2.35
 - (C) 3.25
 - (D) 2.25
56. Picking a card at random from cards numbered 1 to 10, the probability of getting an even number is:
- (A) $\frac{1}{2}$
 - (B) 5
 - (C) $\frac{1}{10}$
 - (D) $\frac{2}{10}$

57. In cold weather a woollen sweater keeps the body warm chiefly because its fibres:
- (A) trap air, which is a poor conductor of heat
 - (B) produce heat by themselves
 - (C) reflect all body heat like a mirror
 - (D) are good conductors of heat
58. A leaf has about 60 000 stomata in all, and 5 out of every 6 are on its lower surface. How many stomata are on the lower surface?
- (A) 12 000
 - (B) 50 000
 - (C) 55 000
 - (D) 10 000
59. The fraction $18/24$ in its simplest (lowest) form is:
- (A) $2/3$
 - (B) $3/4$
 - (C) $9/12$
 - (D) $6/8$
60. If an outcome is absolutely sure to occur, what value is assigned to its probability?
- (A) 1
 - (B) $1/2$
 - (C) 0
 - (D) 100
61. At night a land breeze blows from the land toward the sea because, after sunset, the:
- (A) land and sea cool at the same rate
 - (B) sea cools faster than the land
 - (C) the Sun heats the land at night
 - (D) land cools faster than the sea
62. In a forest survey, $3/5$ of the plants were green food-makers and the rest were fungi. What fraction were fungi?
- (A) $3/2$
 - (B) $3/5$
 - (C) $1/5$
 - (D) $2/5$
63. A plant grows 0.8 cm each day. In 5 days, ignoring nothing, it grows:
- (A) 0.4 cm
 - (B) 4.0 cm
 - (C) 4.5 cm
 - (D) 40 cm

64. Eight students scored 6, 7, 5, 8, 6, 9, 7, 8 on a quiz. The mean score is:
- (A) 7
 - (B) 8
 - (C) 6.5
 - (D) 56
65. One end of an iron rod is held in a flame. The far end becomes hot too because heat is passed along by the:
- (A) light shining from the flame
 - (B) air currents inside the metal
 - (C) particles physically travelling to the far end
 - (D) vibrating particles passing energy to their neighbours
66. Why are most stomata found on the LOWER surface of a leaf rather than the upper?
- (A) to let rain fall straight into them
 - (B) to reduce water loss, since the lower surface is cooler and out of direct Sun
 - (C) because the upper surface has no cells
 - (D) to catch more sunlight for photosynthesis
67. Half of one-third of a chocolate bar is what fraction of the whole bar?
- (A) $\frac{2}{3}$
 - (B) $\frac{1}{6}$
 - (C) $\frac{1}{3}$
 - (D) $\frac{1}{5}$
68. The double bar graph shows boys (14) and girls (16) who like science. Together, the number who like science is:
- (A) 14
 - (B) 2
 - (C) 30
 - (D) 16
69. On a sunny day a dark-coloured car gets hotter inside than a white one because dark surfaces:
- (A) absorb more heat than light surfaces
 - (B) are always made of thicker metal
 - (C) reflect more heat than light surfaces
 - (D) let heat pass straight through
70. The three things, besides chlorophyll, that a green leaf needs to carry out photosynthesis are:
- (A) oxygen, soil and warmth
 - (B) nitrogen, starch and air
 - (C) sunlight, oxygen and salt
 - (D) sunlight, water and carbon dioxide

71. What is 0.125 written as a fraction in lowest terms?

- (A) $\frac{1}{4}$
- (B) $\frac{125}{100}$
- (C) $\frac{1}{8}$
- (D) $\frac{1}{5}$

72. In which situation is the MEDIAN a better 'typical' value than the mean?

- (A) when the data has no numbers at all
- (B) when there are only two values
- (C) when one or two values are extremely high or low compared with the rest
- (D) when all the values are exactly equal

73. A laboratory thermometer differs from a clinical one mainly in that the laboratory thermometer:

- (A) shows temperature in metres
- (B) reads a much wider range, often -10°C to 110°C
- (C) does not use any liquid
- (D) can only measure body heat

74. Insectivorous plants still have green leaves and can photosynthesise. They trap insects only because their soil is poor in:

- (A) sunlight
- (B) carbon dioxide
- (C) nitrogen
- (D) water

75. Comparing $\frac{5}{8}$ and $\frac{3}{5}$, which is larger?

- (A) cannot be compared
- (B) $\frac{3}{5}$
- (C) they are equal
- (D) $\frac{5}{8}$

76. Raw data simply collected, before it is arranged or organised, is called:

- (A) a frequency table
- (B) raw data
- (C) a bar graph
- (D) the mean

77. In physics, heat is most correctly classified as a form of what?

- (A) a force
- (B) energy
- (C) temperature
- (D) matter

78. Oxygen is released by green plants as a by-product of:
- (A) respiration
 - (B) transpiration
 - (C) digestion in roots
 - (D) photosynthesis
79. What is $1 - (2/5 + 1/10)$?
- (A) $3/10$
 - (B) $2/5$
 - (C) $7/10$
 - (D) $1/2$
80. A frequency table is used to show:
- (A) only the largest value
 - (B) how many times each value or category occurs
 - (C) the average of all the values
 - (D) the order in which data was collected
81. Sea coasts have a milder climate than inland places largely because water, compared with land:
- (A) heats up and cools down more slowly
 - (B) cannot absorb any heat
 - (C) blocks all wind from the land
 - (D) heats up and cools down faster
82. Why must non-green parasitic plants like *Cuscuta* attach to a host?
- (A) the host gives them sunlight
 - (B) they grow too tall to stand alone
 - (C) they need extra water only from the host
 - (D) they lack chlorophyll, so they cannot make their own food
83. A recipe needs $2\frac{1}{2}$ cups of flour, but you make half the recipe. How much flour do you need?
- (A) 5 cups
 - (B) 2 cups
 - (C) $1\frac{1}{2}$ cups
 - (D) $1\frac{1}{4}$ cups
84. Ten daily maximum temperatures ($^{\circ}\text{C}$) had a total of 360. Their mean maximum temperature was:
- (A) 10°C
 - (B) 36°C
 - (C) 360°C
 - (D) 3.6°C

85. On five nights the lowest temperatures were 12, 9, 15, 9 and 10 °C. The range of these night readings is:

- (A) 6 °C
- (B) 9 °C
- (C) 15 °C
- (D) 11 °C

86. A class lists what each named plant eats. Cuscuta on a host, bread mould on dead bread, and a mango tree in the Sun are, in order:

- (A) autotroph, parasite, saprotroph
- (B) parasite, autotroph, saprotroph
- (C) parasite, saprotroph, autotroph
- (D) saprotroph, parasite, autotroph

87. The decimal 3.07 means:

- (A) 3 ones and 7 tenths
- (B) 30 ones and 7 tenths
- (C) 3 ones, 0 tenths and 7 hundredths
- (D) 3 ones and 7 ones

88. Picking a day at random from a normal week, the probability that it is a weekend day (Saturday or Sunday) is:

- (A) $\frac{5}{7}$
- (B) $\frac{1}{7}$
- (C) $\frac{1}{2}$
- (D) $\frac{2}{7}$

89. A thermos flask keeps drinks hot for hours mainly because the vacuum between its double walls:

- (A) lets air circulate to keep it warm
- (B) adds heat energy to the drink
- (C) speeds up heat loss by radiation
- (D) stops heat loss by conduction and convection

90. The raw material that a plant draws up from the SOIL for photosynthesis, along with minerals, is:

- (A) oxygen
- (B) carbon dioxide
- (C) water
- (D) sunlight

91. What is $\frac{7}{8} - \frac{1}{2}$?

- (A) $\frac{1}{2}$
- (B) $\frac{6}{6}$
- (C) $\frac{4}{8}$
- (D) $\frac{3}{8}$

- 92.** The mean of four numbers is 15. Their total must be:
- (A) 15
 - (B) 60
 - (C) 3.75
 - (D) 19
- 93.** Why do we often wear loose, light cotton clothes in summer rather than tight ones?
- (A) they let air move over the skin so sweat evaporates and cools us
 - (B) cotton produces cool air by itself
 - (C) they trap a thick layer of warm air
 - (D) they reflect all the body's heat back in
- 94.** Why are green plants called the 'producers' in a food chain?
- (A) they live the longest of all organisms
 - (B) they eat the remains of dead animals
 - (C) they make food that all other organisms ultimately depend on
 - (D) they are the largest living things
- 95.** Multiplying a number by 0.1 has the same effect as:
- (A) dividing it by 100
 - (B) multiplying it by 10
 - (C) dividing it by 10
 - (D) adding 0.1 to it
- 96.** Why must a bar graph's bars all have the same width and equal gaps between them?
- (A) because the bars must touch each other
 - (B) so that only the heights, which show the values, are compared fairly
 - (C) so each bar can show a different unit
 - (D) to make the graph look colourful and neat
- 97.** Why is the bulb of a clinical thermometer made small and the bore (tube) very narrow?
- (A) so it can measure very high temperatures
 - (B) so the glass does not break in heat
 - (C) so even a small heat change makes the thread move a lot, giving a sensitive reading
 - (D) so it can hold a very large amount of liquid
- 98.** The tiny green structures inside leaf cells that contain chlorophyll and where photosynthesis takes place are the:
- (A) stomata
 - (B) chloroplasts
 - (C) veins
 - (D) root hairs

99. A glass is $\frac{3}{5}$ full of water. To fill it completely, the fraction of the glass still to be added is:

- (A) $\frac{3}{5}$
- (B) $\frac{2}{5}$
- (C) $\frac{8}{5}$
- (D) $\frac{1}{5}$

100. The most common value, the middle value of ordered data, and the sum-divided-by-count of a data set are called, in that order, the:

- (A) mean, mode and median
- (B) median, mean and mode
- (C) range, mode and mean
- (D) mode, median and mean